



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**SENIOR CERTIFICATE EXAMINATIONS/
NATIONAL SENIOR CERTIFICATE EXAMINATIONS/
SENIORSERTIFIKAAT-EKSAMEN/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN**

TECHNICAL SCIENCES P1/TEGNIESE WETENSKAPPE V1

MAY/JUNE/MEI/JUNIE 2026

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

**These marking guidelines consist of 13 pages./
Hierdie nasienriglyne bestaan uit 13 bladsye.**

QUESTION/VRAAG 1

1.1	A	✓✓	(2)
1.2	B	✓✓	(2)
1.3	A	✓✓	(2)
1.4	C	✓✓	(2)
1.5	B	✓✓	(2)
1.6	D	✓✓	(2)
1.7	B	✓✓	(2)
1.8	D	✓✓	(2)
1.9	D	✓✓	(2)
1.10	A / B	✓✓	(2)
			[20]

QUESTION/VRAAG 2

- 2.1 A single force which can produce the same effect as two or more forces. ✓✓/
'n Enkele krag wat dieselfde effek as twee of meer kragte teweeg kan bring.

OR/OF

A single force that has the same effect as the combined effects of all forces acting at a point./'n Enkele krag wat dieselfde effek het as die gekombineerde effekte van al die kragte wat op 'n punt inwerk.

(2)

2.2

OPTION/OPSIE 1	OPTION/OPSIE 2

ACCEPTABLE LABELS/ AANVAARBARE ETIKETTE	NOTES/NOTAS
N/F_N: Normal/<i>Normaal</i> F_g/w: Force due to gravity/weight/ <i>Krag as gevolg van gravitasie/gewig</i> $F_{\text{engine/enjin}}/F_A/F$: Force applied by engine/<i>Krag toegepas deur enjin</i>/1 236 N $T//F_T$/Tension/<i>Spanning</i>/$F_{\text{boy/seun}}$/ Force in rope/<i>Krag in tou</i> $f_k/f/F_f$: Kinetic friction/<i>Kinetiese</i> wrywing/<i>Frictional force</i>/ <i>Wrywingskrag</i>	ONE mark for each force represented by an arrow with a correct label./<i>EEN punt vir elke krag verteenwoordig deur 'n pyl met 'n korrekte byskrif.</i> <u>Penalise ONCE for each of the following/Penaliseer EEN KEER vir elk van die volgende:</u> • No arrows/<i>Geen pyltjies</i> • No dot/<i>Geen punt</i> • Gap between the line and the dot/<i>Gaping tussen die lyn en punt</i> • Dotted lines used/<i>Gebroke lyn word gebruik</i> • A force diagram given/'n <i>Kragte diagram word gegee</i> • Additional forces are given/<i>Addisionele kragte word gegee</i>

(5)

2.3.1 $F_{\text{net/netto}} = ma \checkmark$
 $= (1\,800 \times 0,05) \checkmark$
 $= 90 \text{ N} \checkmark$

(3)

2.3.2 **POSITIVE MARKING FROM QUESTION 2.3.1/POSITIEWE NASIEN VANAF VRAAG 2.3.1**

OPTION/OPSIE 1 (Right is positive/Regs is positief)	
$F_{\text{net/netto}} = ma$ $F_{\text{net/netto}} = (+F_{\text{engine/enjin}}) + (-f_k) + (-T \sin \theta)$ $90 \checkmark = (+1236) - 500 - T \sin 79^\circ \checkmark$ $T = 658,09 \text{ N} \checkmark$	$\checkmark$ Any one/Enige een
OPTION/OPSIE 2 (Right is negative/Regs is negatief)	
$F_{\text{net/netto}} = ma$ $F_{\text{net/netto}} = (-F_{\text{engine/enjin}}) + (f_k) + (T \sin \theta)$ $-90 \checkmark = (-1236) + 500 + T \sin 79^\circ \checkmark$ $T = -658,09 \text{ N}$ $= 658,09 \text{ N} \checkmark$	$\checkmark$ Any one/Enige een

OPTION/OPSIE 3 (Right is positive/Regs is positief)	
$F_{\text{net/netto}} = ma$ $F_{\text{net/netto}} = (+F_{\text{engine/enjin}}) + (-f_k) + (-T \cos \theta)$ $90 \checkmark = (+1236) - 500 - T \cos 11^\circ \checkmark$ $T = 658,09 \text{ N} \checkmark$	$\checkmark$ Any one/Enige een
OPTION/OPSIE 4 (Right is negative/Regs is negatief)	
$F_{\text{net/netto}} = ma$ $F_{\text{net/netto}} = (-F_{\text{engine/enjin}}) + f_k + (T \cos \theta)$ $-90 \checkmark = (-1236) + 500 + T \cos 11^\circ \checkmark$ $T = -658,09 \text{ N}$ $= 658,09 \text{ N} \checkmark$	$\checkmark$ Any one/Enige een

(4)

2.4.1 Weight/Gravitational force $\checkmark$ /Gewig/Gravitasiekrag
 The force of attraction the earth exerts on any object on or near its surface. $\checkmark \checkmark$ /
 Die aantrekkingskrag wat die aarde uitoefen op enige voorwerp op of naby sy oppervlak.

(3)

2.4.2 When the boy hits the trampoline, the trampoline simultaneously exerts an oppositely directed force $\checkmark$ of equal magnitude $\checkmark$ on the boy. /Wanneer die seun die trampolien tref, oefen die trampolien gelyktydig 'n teenoorgestelde krag van gelyke grootte op die seun uit.

(2)

2.4.3 Newton's Third Law of motion $\checkmark$ /Newton se Derde Bewegingswet
 When body **A** exerts a force on body **B**, body **B** simultaneously exerts an oppositely directed force of the same magnitude on body **A**. $\checkmark \checkmark$ /Wanneer liggaam **A** 'n krag op liggaam **B** uitoefen, oefen liggaam **B** gelyktydig 'n krag gelyk in grootte en teenoorgesteld in rigting op liggaam **A** uit.

(3)

[22]

QUESTION/VRAAG 3

- 3.1 Vector quantity ✓/Vektorhoeveelheid
It has both magnitude and direction. ✓✓/Dit het beide grootte en rigting. (3)

3.2	OPTION/OPSIE 1 (Right is positive/Regs as positief)	OPTION/OPSIE 2 (Left is positive/Links as positief)	
	$\sum p_{(i)} = m_{AVA} + m_{BVB} \checkmark$ $= (3)(5) \checkmark + (5)(-2) \checkmark$ $= 5 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1} \text{ right/regs} \checkmark$	$\sum p_{(i)} = m_{AVA} + m_{BVB} \checkmark$ $= (3)(-5) \checkmark + (5)(2) \checkmark$ $= -5 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$ $= 5 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1} \text{ right/regs} \checkmark$	(4)

- 3.3 The total linear momentum of an isolated system remains constant ✓ (is conserved) in magnitude and direction. ✓/Die totale lineêre momentum van 'n geïsoleerde sisteem bly konstant (word behou) in grootte en rigting. (2)

3.4 **POSITIVE MARKING FROM QUESTION 3.2/POSITIEWE NASIEN VANAF VRAAG 3.2**

OPTION/OPSIE 1 (Right is positive/Regs is positief)	OPTION/OPSIE 2 (Left is positive/Links is positief)	
$\sum p_{(i)} = \sum p_{(f)} \checkmark$ $(m_{AVA})_{(i)} + (m_{BVB})_{(i)} = (m_A + m_B)v_f$ $5 \checkmark = (m_A + m_B)v_f$ $5 = (3 + 5)v_f \checkmark$ $v_f = 0,625 \text{ m}\cdot\text{s}^{-1} \checkmark$	$\sum p_{(i)} = \sum p_{(f)} \checkmark$ $(m_{AVA})_{(i)} + (m_{BVB})_{(i)} = (m_A + m_B)v_f$ $5 \checkmark = (m_A + m_B)v_f$ $-5 = (3 + 5)v_f \checkmark$ $v_f = -0,625$ $v_f = 0,625 \text{ m}\cdot\text{s}^{-1} \checkmark$	Any one/ Enige een
OPTION/OPSIE 3 (Right is positive/Regs is positief)	OPTION/OPSIE 4 (Left is positive/Links is positief)	
$\sum p_{(i)} = \sum p_{(f)} \checkmark$ $(m_{AVA})_{(i)} + (m_{BVB})_{(i)} = (m_A + m_B)v_f$ $(3)(5) + (5)(-2) \checkmark = (3 + 5)v_f \checkmark$ $v_f = 0,625 \text{ m}\cdot\text{s}^{-1} \checkmark$	$\sum p_{(i)} = \sum p_{(f)} \checkmark$ $(m_{AVA})_{(i)} + (m_{BVB})_{(i)} = (m_A + m_B)v_f$ $(3)(-5) + (5)(2) \checkmark = (3 + 5)v_f \checkmark$ $v_f = -0,625$ $v_f = 0,625 \text{ m}\cdot\text{s}^{-1} \checkmark$	Any one/ Enige een

- 3.5 During a collision/Tydens 'n botsing:
- Contact time increases ✓ as the crumple zone compresses/Kontaktyd neem toe soos die frommelsone saamgedruk word
 - Net force experienced is reduced ✓/Netto krag wat ondervind word neem af
 - Change in momentum remains constant ✓/Verandering in momentum bly konstant

(3)
[16]

QUESTION/VRAAG 4

- 4.1 The force that acts between two surfaces when the object is moving. ✓✓/Die krag wat uitgeoefen word tussen twee oppervlakke indien die voorwerp in beweging is.

OR/OF

The force that opposes the motion of a moving object relative to the surface./ Die krag wat die beweging van 'n voorwerp teenwerk relatief tot die oppervlak.

(2)

4.2



ACCEPTABLE LABELS/ AANVAARBARE ETIKETTE	NOTES/NOTAS
F_A : $F_{\text{applied/toegepas}}$ /15 N $f_k/f/F_f$: Kinetic friction/ <i>Kinetiese wrywing</i> /Frictional force/ <i>Wrywingskrag</i>	ONE mark for each force represented by an arrow with a correct label./EEN punt vir elke krag verteenwoordig deur 'n pyl met 'n korrekte byskrif. Penalise ONCE for each of the following/<i>Penaliseer EEN KEER vir elk van die volgende</i>: - No arrows/<i>Geen pyltjies</i> - No dot/<i>Geen punt</i> - Gap between the line and the dot/<i>Gaping tussen die lyn en punt</i> - Dotted lines used/<i>Gebroke lyn word gebruik</i> - A force diagram given/'n <i>Kragte-diagram word gegee</i> - Additional forces are given/<i>Addisionele kragte word gegee</i>

(2)

4.3.1

OPTION/OPSIE 1	OPTION/OPSIE 2
$f_k = \mu_k N$ ✓ $= (0,15)(4)(9,8)$ ✓ $= 5,88 \text{ N left/links}$ ✓	$N = mg$ $= (4)(9,8)$ $= 39,20 \text{ N}$ $f_k = \mu_k N$ ✓ $= (0,15)(39,2)$ ✓ $= 5,88 \text{ N left/links}$ ✓

(3)

4.3.2

POSITIVE MARKING FROM QUESTION 4.3.1/POSITIEWE NASIEN VANAF VRAAG 4.3.1

OPTION/OPSIE 1	OPTION/OPSIE 2
$W_{\text{net/netto}} = W_{F_A} + W_{f_k}$ ✓ $W_{\text{net/netto}} = F_A \Delta x \cos \theta + f_k \Delta x \cos \theta$ $= (15)(3) \cos 0^\circ + (5,88)(3) \cos 180^\circ$ ✓ $= 27,36 \text{ J}$ ✓	$W_{\text{net/netto}} = F_{\text{net/netto}} \Delta x \cos \theta$ ✓ $W_{\text{net/netto}} = (F_A - f_k) \Delta x \cos \theta$ $= (15 - 5,88) (3) (\cos 0^\circ)$ ✓ $= 27,36 \text{ J}$ ✓

(4)

4.4 Principle of conservation of mechanical energy ✓/Beginnel van behoud van meganiese energie (1)

4.5 $M_E = E_k + E_p$
 $M_E = \frac{1}{2}mv^2 + mgh$ } ✓ Any correct one/Enige korrekte een
 $0,27 = \frac{1}{2}(m)(0)^2 + (m)(9,8)(2,8)$ ✓
 $m = 9,84 \times 10^{-3} \text{ kg}$ ✓ (3)

4.6.1 Increases ✓/Neem toe (1)

4.6.2 **POSITIVE MARKING FROM QUESTION 4.5/POSITIEWE NASIEN VANAF VRAAG 4.5**

E_p at the point when hitting the toolbox/ E_p by die punt wanneer dit die gereedskapkas tref

$$\begin{aligned} E_p &= mgh \checkmark \\ &= (0,00984)(9,8)(1,45) \checkmark \\ &= 0,01398 \text{ J} \checkmark \end{aligned}$$

RANGE/REEKS: 0,1398 J – 0,1421 J

E_p at highest point after bouncing/ E_p by hoogste punt nadat dit bons

$$\begin{aligned} E_p &= mgh \\ &= (0,00984)(9,8)(2,8) \checkmark \\ &= 0,27 \text{ J} \checkmark \end{aligned}$$

E_p at 1,45 m (top of toolbox/) is smaller than E_p at highest point. ✓ /
 E_p by 1,45 m (bokant van gereedskapkas) is kleiner as E_p by hoogste punt.

OR/OF

E_p at highest point is greater than E_p at 1,45 m (top of toolbox)./ E_p by hoogste punt is groter as E_p by 1,45 m (bokant van gereedskapkas). (6)

4.7 **POSITIVE MARKING FROM QUESTION 4.6.1 AND 4.6.2/POSITIEWE NASIEN VANAF VRAAG 4.6.1 EN 4.6.2**

$$0,27 \text{ J} \checkmark$$

(1)
[23]

QUESTION/VRAAG 5

- 5.1 Within the limit of elasticity ✓, stress is directly proportional to the strain. ✓/In die grens van elastisiteit, is spanning direk eweredig aan die rekking. (2)

5.2.1	OPTION/OPSIE 1	OPTION/OPSIE 2
	$\cos 4^\circ = \frac{10}{\frac{1}{2} L_{f \text{ rope/tou}}}$ $\frac{1}{2} L_{f \text{ rope/tou}} = \frac{10}{\cos 4^\circ} \checkmark$ $= 10,0244 \text{ m}$ $\Delta \ell = L_f - L_i$ $\Delta \ell = (10,0244 \times 2) \checkmark - 20 \checkmark$ $\Delta \ell = 0,05 \text{ m} \checkmark$	$\sin 86^\circ = \frac{10}{\frac{1}{2} L_{f \text{ rope/tou}}}$ $\frac{1}{2} L_{f \text{ rope/tou}} = \frac{10}{\sin 86^\circ} \checkmark$ $= 10,0244 \text{ m}$ $\Delta \ell = L_f - L_i$ $\Delta \ell = (10,0244 \times 2) \checkmark - 20 \checkmark$ $\Delta \ell = 0,05 \text{ m} \checkmark$
	RANGE/REEKS: 0,04 m - 0,05 m	

- 5.2.2 **POSITIVE MARKING FROM QUESTION 5.2.1/POSITIEWE NASIEN VANAF VRAAG 5.2.1**

OPTION/OPSIE 1	OPTION/OPSIE 2
$\varepsilon = \frac{\Delta \ell}{L} \checkmark$ $= \frac{0,05}{20} \checkmark$ $= 2,50 \times 10^{-3} \checkmark$	$\varepsilon = \frac{\Delta \ell}{L} \checkmark$ $= \frac{0,0488}{20} \checkmark$ $= 2,44 \times 10^{-3} \checkmark$

RANGE/REEKS: $2,44 \times 10^{-3} - 2,50 \times 10^{-3}$ (3)

- 5.3
- Quartz fire in guitar strings ✓/Kwartsvuur in kitaarsnare
 - Phosphorus bronze in springs and cables ✓/Fosforbrons in vere en kables
 - Elastic/rubber band/Elastiese/rubberband
 - Bow and arrow/Pyl en boog (Any correct two/Enige korrekte twee) (2)

[11]

QUESTION/VRAAG 6

- 6.1 The normal force exerted by a liquid at rest on a given surface in contact with it. ✓✓/Die normaalkrag wat uitgeoefen word deur 'n vloeistof in rus met 'n gegewe oppervlak wat daarmee in kontak is. (2)

OPTION/OPSIE 1	OPTION/OPSIE 2
$A = \pi r^2$ $A = \pi(0,28)^2 \checkmark$ $A = 0,2463 \text{ m}^2$ $\frac{F_1}{A_1} = \frac{F_2}{A_2} \checkmark$ $\frac{350}{0,02} \checkmark = \frac{F_2}{0,2463} \checkmark$ $F_2 = 4\,310,25 \text{ N} \checkmark$	$\frac{F_1}{A_1} = \frac{F_2}{A_2} \checkmark$ $\frac{350}{0,02} \checkmark = \frac{F_2}{\pi(0,28)^2} \checkmark$ } ✓ Substitution / Substitusie $F_2 = 4\,310,27 \text{ N} \checkmark$
RANGE/REEKS: 4 305,00 N - 4 375 N	

OPTION/OPSIE 3
$A = \frac{\pi d^2}{4}$ $A = \frac{\pi(560 \times 10^{-3})^2}{4} \checkmark$ $A = 0,2463 \text{ m}^2$ $\frac{F_1}{A_1} = \frac{F_2}{A_2} \checkmark$ $\frac{350}{0,02} \checkmark = \frac{F_2}{0,2463} \checkmark$ $F_2 = 4\,310,25 \text{ N} \checkmark$
RANGE/REEKS: 4 305,00 N - 4 375 N

- 6.3.1 Remains the same ✓/Bly dieselfde (1)
- 6.3.2 Pressure depends only on force applied and the area. ✓✓/Druk is slegs afhanklik van die toegepaste krag en die area.

OR/OF

Pressure is independent of the length of the cylinder./Druk is onafhanklik van die lengte van die silinder. (2)

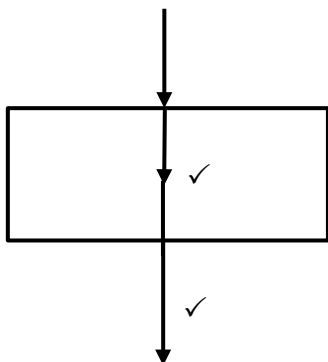
- 6.4
- Vehicle lifts ✓/Voertuighysers
 - Forklifts/Vurkhyzers
 - Dentist's chair/Tandarts se stoel
 - Hydraulic jack/Hidrouliese domkrag
 - Hydraulic brakes/Hidrouliese remme (Any correct one/Enige korrekte een) (1)

[11]

QUESTION/VRAAG 7

- 7.1 The bending of light ✓ when it travels from one medium to another ✓ (of different optical density). /Die buiging van lig wanneer dit vanaf een medium na 'n ander (met verskillende optiese digtheid) beweeg. (2)

7.2

**MARKING CRITERIA/****NASIENKRITERIA**

- Ray passes through glass block without bending/Straal beweeg deur glasblok sonder buiging (1)
- Ray emerges the glass block without bending/Straal verlaat die glasblok sonder buiging (1)

(2)

- 7.3.1 Refracted ray ✓ /Brekingstraal (1)

- 7.3.2 Normal (line) ✓ /Normaal(lyn) (1)

- 7.4 Glass ✓ /Glas (1)

7.5 **NEGATIVE MARKING FROM QUESTION 7.4/NEGATIEWE NASIEN VANAF VRAAG 7.4**

When a light ray travels from a medium of higher optical density to that of a lower optical density ✓ it bends away from the normal, as it is in the case of ray 1. ✓ /Wanneer 'n ligstraal vanaf 'n medium met hoër optiese digtheid na daardie met 'n laer optiese digtheid beweeg, buig dit weg vanaf die normaal, soos in die geval van straal 1.

OR/OF

Total internal reflection occurs in the denser medium, as it is in the case of ray 3. /Totale interne weerkaatsing vind plaas in die digter medium, soos in die geval van straal 3. (2)

- 7.6.1 Total internal reflection ✓ /Totale interne weerkaatsing (1)

- 7.6.2
- The angle of incidence must be greater than the critical angle ✓ /Die invalshoek moet groter wees as die grenshoek.
 - Light must move from a medium of higher to lower optical density ✓ /Lig moet vanaf 'n medium met hoër na laer optiese digtheid beweeg. (2)

- 7.7.1 B ✓ (Remote controllers; photography and thermal imaging/ Afstandbeheerders; fotografie en termiese beelding). (1)

- 7.7.2 A ✓ (Used to treat cancer; sterilise food and instruments/Gebruik om kanker te behandel; sterilisasie van voedsel en instrumente). (1)

7.8 $E = \frac{hc}{\lambda}$ ✓
 $1 \times 10^{-14} \text{ ✓} = \frac{(6,63 \times 10^{-34})(3 \times 10^8)}{\lambda}$ ✓
 $\lambda = 1,989 \times 10^{-11} \text{ m ✓}$

RANGE/REEKS: $1,989 \times 10^{-11} \text{ m}$ to $2 \times 10^{-11} \text{ m}$

(4)

- 7.9.1
- Convex lens is thicker in the middle and thinner at the edges. ✓/Konvekse lens is dikker in die middel en dunner op die kante.
 - Concave lens is thinner in the middle and thicker at the edges. ✓/Konkawe lens is dunner in die middel en dikker op die kante.

OR/OF

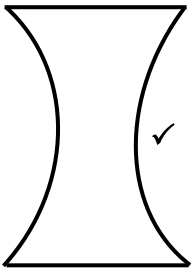
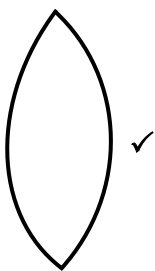
- Convex lens converges light rays./Konvekse lens konvergeer ligstrale.
- Concave lens diverges light rays./Konkawe lens divergeer ligstrale.

OR/OF

- Convex lens gives a real and virtual image./Konvekse lens gee ware en skynbeeld.
- Concave lens gives a virtual image only./Konkawe beeld gee slegs 'n skynbeeld.

(2)

7.9.2

Concave/Konkawe lens	Convex/Konvekse lens
	

(2)

[22]

QUESTION/VRAAG 8

- 8.1 A device that stores electrical charge. ✓✓/’n Toestel wat elektriese lading stoor. (2)

8.2 $C = \frac{\epsilon_0 A}{d}$
 $= \frac{(8,85 \times 10^{-12})(0,04 \times 0,04)}{0,0025}$ ✓

$= 5,664 \times 10^{-12} \text{ F}$

$C = \frac{Q}{V}$ ✓
 $5,664 \times 10^{-12} \text{ ✓} = \frac{5 \times 10^{-11}}{V}$ ✓
 $V = 8,83 \text{ V ✓}$

One mark to any ONE of the two formulae/Een punt vir enige EEN van die twee formules

(5)

[7]

QUESTION/VRAAG 9

- 9.1 Opposition to the flow of electric current. ✓✓/Teenstand teen die vloeï van elektriese stroom. (2)

9.2.1

OPTION/OPSIE 1	OPTION/OPSIE 2
$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} \checkmark$ $= \frac{1}{4} + \frac{1}{4} \checkmark$ $R_p = 2 \Omega \checkmark$	$R_p = \frac{R_1 \times R_2}{R_1 + R_2} \checkmark$ $= \frac{4 \times 4}{4 + 4} \checkmark$ $= 2 \Omega \checkmark$

(3)

9.2.2

OPTION/OPSIE 1	OPTION/OPSIE 2
$W = I^2 R \Delta t \checkmark$ $= (1,2)^2 (3)(3) \checkmark$ $= 12,96 \text{ J} \checkmark$	$V = IR$ $= 1,2 \times 3$ $= 3,6 \text{ V}$ $W = VI \Delta t \checkmark$ $= (3,6)(1,2)(3) \checkmark$ $= 12,96 \text{ J} \checkmark$
OPTION/OPSIE 3	OPTION/OPSIE 4
$V = IR$ $= 1,2 \times 3$ $= 3,6 \text{ V}$ $W = \frac{V^2 \Delta t}{R} \checkmark$ $= \frac{(3,6)^2 (3)}{3} \checkmark$ $= 12,96 \text{ J} \checkmark$	$Q = I \Delta t$ $= 1,2 \times 3$ $= 3,6 \text{ C}$ $W = VQ \checkmark$ $= 3,6 \times 3,6 \checkmark$ $= 12,96 \text{ J} \checkmark$

(3)

- 9.3 Decrease ✓/Afneem

(1)

[9]**QUESTION/VRAAG 10**

- 10.1 The process of generating electricity from motion. ✓✓ /Die proses om elektrisiteit deur beweging op te wek.

OR/OF

The production of an electromotive force (emf) or voltage across an electrical conductor due to relative motion between the conductor and magnetic field./ Die produksie van 'n elektromotoriese krag (emk) of spanning oor 'n elektriese geleier a.g.v. relatiewe beweging tussen die geleier en magneetveld. (2)

10.2

$$\varepsilon = - \frac{N \Delta \Phi}{\Delta t} \checkmark$$

$$0,25 \checkmark = \frac{-250 (1 \times 10^{-4} - 5 \times 10^{-4})}{\Delta t} \checkmark$$

$$\Delta t = 0,4 \text{ s} \checkmark$$

(4)

- | | | |
|--------|--|-----|
| 10.3 | Ensures that current flows in one direction. ✓/Verseker dat stroom in een rigting vloeï. | (1) |
| 10.4.1 | Anticlockwise ✓/Antikloksgewys | (1) |
| 10.4.2 | Fleming's left-hand (motor) rule ✓/Fleming se linkerhand(motor)reël | (1) |

[9]

TOTAL/TOTAAL: 150